Year 13 Further Maths - Modelling with power

Similar questions - Cars, lorries and variable resistance

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A van of mass 1500 kg has a constant total resistance of 600 N.

(a) The engine works at 12 kW. Find the acceleration when the speed is 8 $\mathrm{m\,s^{-1}}$ on a horizontal road. **[3]**

(b) The maximum power of the engine is 21 kW. Find the greatest constant speed on a horizontal road. **[2]**

(c) The van now travels up a road inclined at θ to the horizontal, where $\sin\theta = 0.04$. Find the greatest constant speed up the hill, giving your answer to 3 s.f. **[3]**

2 A lorry of mass 8000 kg has maximum engine power 120 kW. On a horizontal road the greatest constant speed is 30 $\mathrm{m\,s^{-1}}$. Assume the resistance is constant.

(a) Show that the resistance is 4000 N. **[2]**

(b) Find the acceleration when the lorry travels at 15 $\mathrm{m\,s^{-1}}$ with power 80 kW. **[3]**

(c) Driving down a slope inclined at θ to the horizontal at maximum power, the greatest constant speed is 40 $\mathrm{m\,s^{-1}}$. Find θ in degrees to 2 s.f. **[4]**

3 A car of mass 1000 kg has maximum power 20 kW. The resistance to motion is modelled as kv newtons at speed v $\mathrm{m\,s^{-1}}$ where k is a constant. The maximum speed on a horizontal road is 40 $\mathrm{m\,s^{-1}}$.

(a) Show that $k = 12.5$. **[3]**

(b) Find the acceleration at 20 $\mathrm{m\,s^{-1}}$ when the engine is working at 20 kW. **[3]**

(c) Find the power required to maintain a constant speed of 30 $\mathrm{m\,s^{-1}}$ on a horizontal road. **[2]**

Question 1

(a) Driving force $= \frac{12000}{8} = 1500$ N; $1500 - 600 = 1500a$, $a = 0.6\ \mathrm{m\,s^{-2}}$

(b) $v = \frac{21000}{600} = 35\ \mathrm{m\,s^{-1}}$

(c) Driving force $= 600 + 1500 \times 9.8 \times 0.04 = 1188$ N; $v = \frac{21000}{1188} = 17.7\ \mathrm{m\,s^{-1}}$

Question 2

(a) At constant speed driving force = resistance, so $R = \frac{120000}{30} = 4000$ N

(b) $F = \frac{80000}{15} = 5333\frac{1}{3}$ N; $5333.3 - 4000 = 8000a$, $a = 0.167\ \mathrm{m\,s^{-2}}$

(c) $F = \frac{120000}{40} = 3000$; $3000 + 8000 \times 9.8 \sin\theta = 4000$, $\sin\theta = 0.01276$, $\theta = 0.73^\circ$

Question 3

(a) At max speed driving force $= \frac{20000}{40} = 500 = 40k$, so $k = 12.5$

(b) $F = \frac{20000}{20} = 1000$, $R = 250$; $750 = 1000a$, $a = 0.75\ \mathrm{m\,s^{-2}}$

(c) $F = R = 12.5 \times 30 = 375$ N, so $P = 375 \times 30 = 11250$ W $= 11.25$ kW